

AI作业-机器学习

1. (a) 使用贪心策略, 状态2的最大价值 $V^* = \max\{6, 8, 3\} = 8$

则 $Q(3, left) = reward + \gamma V(2)$

$$= -1 + 0.9 \times 8$$

$$= 6.2$$

(b) 时序差分公式:

$$Q(1, right) = Q(1, right) + \alpha(reward + \gamma Q(2) - Q(1, right)) = 3 + 0.2(-1 + 0.8 \times 8 - 3) = 3.2$$

$$Q(2, up) = Q(2, up) + \alpha(reward + \gamma Q(5) - Q(2, up)) = 8 + 0.2(-1 + 0.8 \times 8 - 8) = 7.2$$

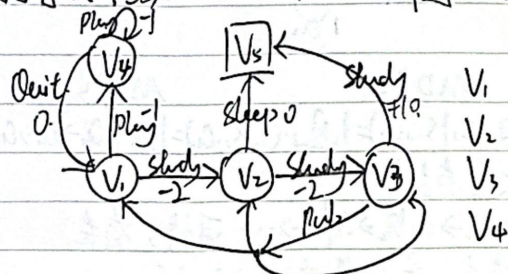
$$Q(5, right) = Q(5, right) + \alpha(reward + \gamma Q(6) - Q(5, right)) = 8 + 0.2(1 + 0 - 8) = 8.4$$

2. (a) 设问号节点值函数为V

$$2.7 \times 0.5 + 0 + 1 \times 0.5 + 0 = 0. V = 0.5 \times (1 + 0) + 0.5 \times (0.4 \times 2.7 + 0.2 \times (-1.3) + 0.4 \times V_{\text{?}})$$

$$2.7 \times 0.5 + 0 + 1 \times 0.5 + 0 = 0. \text{ 求解: } V = \frac{7.4}{1.5} = 4.93$$

(b)



$$Q(V_1, \text{Quit}) = -1.3 + reward + Q(V_4, \cdot) = -1 + 0 = -1$$

$$Q(V_1, \text{Study}) = reward + Q(V_2, \cdot) = -2 + 0 = -2$$

$$Q(V_2, \text{Sleep}) =$$

到达最终状态V5的动作价值函数有: $Q^*(V_2, \text{Sleep}) = 0, Q^*(V_3, \text{Study}) = 1.0$

$$Q^*(V_2, \text{Study}) = Q^*(V_3, \cdot) + reward = 1.0 - 2 = -1$$

$$Q^*(V_3, \text{Quit}) = Q^*(V_2, \cdot) + reward = -1 + (-2) = -3$$

$$Q^*(V_3, \text{Study}) = [0.4 \times Q^*(V_2, \cdot) + 0.4 \times Q^*(V_1, \cdot) + 0.2 \times Q^*(V_3, \cdot)] + reward = 0.4 \times (-1) + 0.4 \times (-2) + 0.2 \times 1.0 + 1.0 = 0.2$$

$$Q^*(V_4, \text{Quit}) = Q^*(V_1, \cdot) + reward = -2 + 0 = -2$$

$$Q^*(V_4, \text{Study}) = Q^*(V_5, \cdot) + reward = 0 + 1 = 1$$

$$Q^*(V_1, \text{Quit}) = Q^*(V_4, \cdot) + reward = 1 - 1 = 0$$

3. MDP:

(1) MDP1:

转移: $P(S_1|S_1, a_1)=1, P(S_2|S_1, a_2)=1, P(S_1|S_2, a_1)=1, P(S_2|S_2, a_2)=1$

奖励: $R(S_2|S_1, a_2)=1$, 否则为0

(2) MDP2:

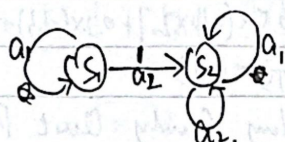
转移: $P(S_1|S_1, a_2)=1, P(S_2|S_1, a_2)=1, P(S_2|S_2, a_1)=1, P(S_2|S_2, a_2)=2$

奖励: $R(S_2|S_2, a_1)=1, R(S_2|S_2, a_2)=1$, 否则为0

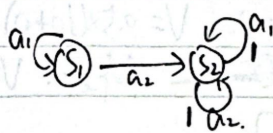
(3) MDP3:

转移: $P(S_1|S_1, a_1)=1, P(S_2|S_1, a_2)=1, P(S_1|S_2, a_1)=1, P(S_2|S_2, a_2)=1$

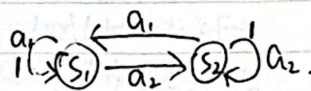
奖励: $R(S_1|S_1, a_1)=1, R(S_2|S_2, a_2)=1$, 否则为0



MDP1



MDP2



MDP3

(a) i) ~~存在~~, 对于MDP3, $R(S_1|S_1, a_1)=1$, 且 $P(S_1|S_1, a_1)=1$, 则 $a_1 \rightarrow a_1 \rightarrow a_1 \dots$ 使得 V_1 值无限增大。存在

ii) 对于MDP2, $S_1 \rightarrow a_2 \rightarrow a_1/a_2 \rightarrow a_1/a_2 \dots$ 可行。存在

iii) 对于MDP1, 不存在。最优策略 $S_1 \rightarrow a_2 \rightarrow S_2$

(b) i) MDP1: $S_1 \rightarrow a_2 \rightarrow S_2$

ii) MDP2: $S_1 \rightarrow a_2 \rightarrow a_1/a_2 \rightarrow a_1/a_2 \dots$

iii) MDP3: $S_1 \rightarrow a_1 \rightarrow a_1 \rightarrow a_1 \dots$